What momentum says about impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect stopping impulse to average force without claiming a complete damage forecast.

SUCCESS CRITERIA - I can...

1. Compute momentum change during a stop.
2. Compare average force for different stopping times.
3. Name information beyond momentum needed for damage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two **3 kg** carts at **+5** and **-3 m/s** stick. **Find final velocity** and use kinetic energy to **explain why this** is not elastic.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $|F_{avg}| = |\Delta p|/\Delta t$.
Stopping force depends on stopping time as well as momentum. These quantities alone do not specify damage.

Worked reasoning: Two incoming objects have equal momentum **12 kg·m/s** but masses **2** and **4 kg**, so their speeds are **6** and **3 m/s**. Momentum alone does not name every possible consequence of stopping them.

Your turn: Draw equal momentum arrows and unequal velocity arrows.

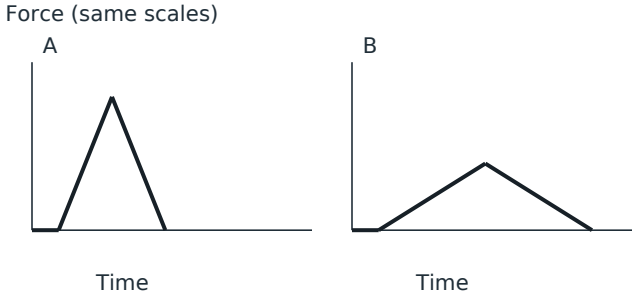


2 kg; 6 m/s 4 kg; 3 m/s

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two equal-momentum objects stop in **0.1 s** and **1 s**. Which has greater average force magnitude, and by what factor?



Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical object of mass **10^6 kg** at **200 m/s** stops in **0.50 s**. **Find impulse** and average force magnitude. Then **compare a 5 s** stop.

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

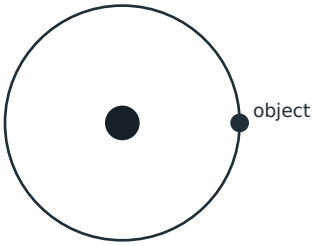
05 Explain your reasoning

Why is a single stopping-force number not a prediction of buildings damaged by an asteroid?

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An ideal circular orbit has GM=**576 m³/s²** and r=**16 m**. **Find speed**, inward acceleration and period; **explain how velocity changes**.
Label velocity and inward acceleration on the circular-path diagram.



Marzano check - circle: ① ② ③

07 Exit check

Objects A and B have equal momentum. A has twice B's mass. **Compare speeds** and kinetic energies, showing why equal momentum need not mean equal energy.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A **0.5 kg** ball arrives at **+10 m/s** and either stops or rebounds at **-6 m/s** in **0.04 s**. **Compare impulse** and average force.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③